EE3032 Introduction to VLSI Design—Homework 2

1. Fig. 1(a) shows an inverter driving a capacitive load C_L. If two input stimuli as shown in Fig. 1(b) are applied, the fall time of the output Vout for the input stimulus a is shorter than that for the input stimulus b. Why? Explain the reasons. (2 points)

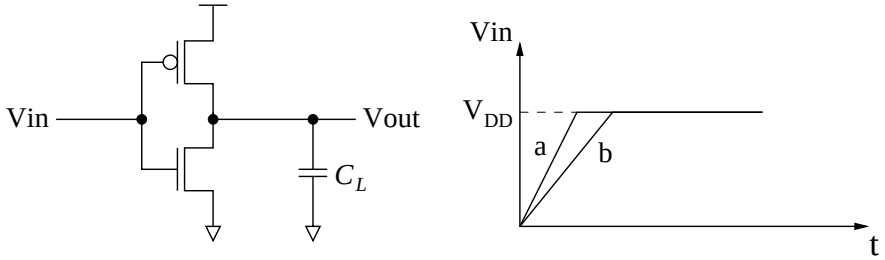

Figure 1: (a) An inverter drives a load C_L. (b) Two input waveforms with different rise times.

2. Fig. 2 shows a simple clock tree. Assume that the delay of a l-μm interconnection can be estimated by $\tau_w = \frac{1}{2}rcl^2$, where r and c denote the unit resistance and capacitance per μm, respectively. Also, assume that $\tau_g = 0.2ns$, $r = 0.05\Omega/\mu m$, and $c = 30 \times 10^{-18}F/\mu m$. If $\tau_w \leq 4\tau_g$, then the delay of the interconnection can be neglected. Which path is the critical path of the clock tree? Calculate the delay of the critical path. (4 points)

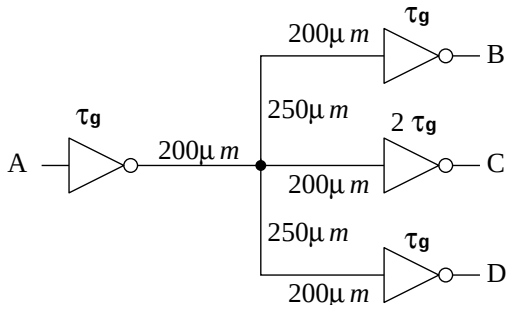

Figure 2: A simple clock tree.

3. Assume that the size of the two transistors of an inverter is as follows: M1=$W_p/L_p = 4$ and M2=$W_n/L_n = 2$. If Vin=$\frac{1}{2}V_{DD}$, the Vout=$\frac{1}{2}V_{DD}$. Calculate the ratio of the μ_n to μ_p. (2 points)

4. Noise margin of a logic gate is defined by two parameters: NM_H (noise margin for logic high) and NM_L (noise margin for logic low). If you desire to design an inverter such that $NM_H > NM_L$, then how do you size the NMOS and PMOS transistors of the inverter? Is $\beta_p > \beta_n$ or $\beta_p < \beta_n$? Explain your answer. (2 points)